Constrained Deep Reinforcement Learning with Lagrangian Theory

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Abstract

Reinforcement learning (RL) agents need to explore their environments in order to learn optimal policies by trial and error. While it is currently typical to train RL agents mostly or entirely in simulation, where safety concerns are minimal. However, it is very difficult to accurately describe complex real-world environments using simulations. Therefore, efforts to train and validate in real-world environments are continuously ongoing. Thus, research on constrained RL has emerged to ensure the stability of agents. In this paper, we propose an algorithm that estimates the Lagrangian multiplier at each step to impose appropriate constraints on each state, ensuring stability.

Contents

1 Introduction **1**
1.1 Background 1
1.2 Research Objectives 1
2 Notations and Preliminaries **1**
2.1 Constrained Markov Decision Processes 2
2.2 State-wise Constrained Markov Decision Process 2
3 Review of Contraction Theory **2**
3.1 Basic Results of Contraction Theory for Deterministic Systems 2
3.1.1 Notable Definitions 3
3.1.2 Fundamental Theorem of Contraction Theory 3
3.1.3 Partial Contraction 4
3.2 Basic Results of Contraction Theory for Stochastic Systems 4
4 Conclusion **4**

1 Introduction

1.1 Background

1.2 Research Objectives

The main objectives of this research are as follows:

- Mathematical stability analysis of the controller and estimator with deep neural networks using the contraction theory.
- Development of the controller and estimator with deep neural networks using the contraction theory.

2 Notations and Preliminaries

An overview of Constrained Markov Decision Processes (CMDP) can be found in this book [1].

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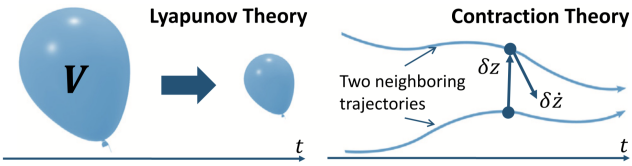

Figure 1: Difference between Lyapunov and contraction theory [3, Fig. 1]. The Lyapunov theory investigates the convergence to a single point and the contraction theory does regarding a single trajectory.

2.1 Constrained Markov Decision Processes

The constrained Markov decision process (CMDP), $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, r, c \rangle$, is to maximize the expected return of rewards while satisfying the constraint on the expected return of costs:

$$\max_{\pi} J(\pi) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty}\gamma^t r_t\right\} \quad \text{s.t. } C(\pi) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty}\gamma_c^t c_t \leq d\right\}$$

where $\mathcal{S}$ and $\mathcal{A}$ are state and action space, π is the stationary policy[1]in stationary policy set Π, $\mathcal{P}$ is the environment transition probability. $r, c : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ are the reward and cost functions, respectively, and γ, γ_C are their discounted factors, $\tau = \{s_0, a_0, s_1, a_1, \cdots\}$ is the trajectory under policy π starting from an initial state distribution ρ. The feasible policy set is defined as follows, where $d_i \in \mathbb{R}$:

$$\Pi_C = \{\pi \in \Pi \mid \forall i, \mathcal{J}_{C_i}(\pi) \leq d_i\}$$

In the definition of a CMDP, constraints are typically formulated in a cumulative or probabilistic form. However, in real-world applications, many critical constraints must be satisfied instantaneously and deterministically.

2.2 State-wise Constrained Markov Decision Process

In this research, we are specifically interested in a special type of CMDP where the safety specification is to persistently satisfy a hard cost constraint at every step, which is called State-wise Constrained Markov Decision Process (SCMDP). SCMDP requires the cost for every state-action transition satisfies a hard constraint. The set of feasible stationary policies for SCMDP is defined as follows:

$$\bar{\Pi}_C = \{\pi \in \Pi \mid \forall (s_t, a_t, s_{t+1}) \sim \tau,\ \forall i,\ C_i(s_t, a_t, s_{t+1}) \leq w_i\}$$

where $\tau \sim \pi$ and $w_i \in \mathbb{R}$. Then, the objective for SCMDP is to find a feasible stationary policy from $\bar{\Pi}_C$ that maximizes the performance measure.

$$\max_{\theta} \mathcal{J}(\pi_\theta), \quad \text{s.t. } \pi_\theta \in \bar{\Pi}_C$$

3 Review of Contraction Theory

For your smooth start, we recommend you to begin with [2]. The overview of contraction theory is presented in a review paper [3].

3.1 Basic Results of Contraction Theory for Deterministic Systems

First, we start with the following deterministic systems:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t), \tag{1}$$

where $\boldsymbol{f}(\boldsymbol{x}, t)$ is an $n \times 1$ sufficiently smooth non-linear vector function and $\boldsymbol{x} \in \mathbb{R}^n$ is the state vector. The smooth property of $\boldsymbol{f}(\boldsymbol{x}, t)$ is essential to ensure the existence and uniqueness of the solution to (1) [4, see, pp. 88-89].

The biggest difference between the traditional Lyapunov theory and the contraction theory is that the contraction theory investigates the convergence of the state trajectory to a single trajectory (contraction behavior),

[1]

1. In CMDP, stationary policies are used to ensure convergence and for mathematical convenience.
2. A stationary policy selects actions based only on the current state, without changing over time, and chooses actions according to the same probability distribution for the same state. Please refer to the book [1] for more details.

while the Lyapunov theory focuses on the convergence of the state trajectory to a single point i.e., see, Fig. 1. For this, motivated by the calculus of variations [5, Chap. 4], (1) can be rewritten as differential dynamics using *differential displacement* $\delta\boldsymbol{x}$ as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}(\boldsymbol{x},t)\delta\boldsymbol{x}. \tag{2}$$

For your information, $\delta\boldsymbol{x}$ is an infinitesimal displacement at *fixed time*.

3.1.1 Notable Definitions

Before we present the fundamental theorem of contraction theory, we introduce the following definitions. One can re-visit this section while reading further.

Definition 3.1 (see, Def. 2.2 [3])**.** If any two trajectories $\boldsymbol{\xi}_1(t)$ and $\boldsymbol{\xi}_2(t)$ of (1) converge to a single trajectory, then the system (1) is said to be *incrementally exponentially stable*, if $\exists C, \alpha > 0$, subject to the following holds:

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| \leq C\|\boldsymbol{\xi}_1(0) - \boldsymbol{\xi}_2(0)\| \exp^{-\alpha t},\ \forall t \in \mathbb{R}_{\geq 0}.$$

The result of Theorem 3.1 equivalently implies the incremental exponential stability, since we have $\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| = \|\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \delta\boldsymbol{x}(t)\|$.

Definition 3.2. Let $\boldsymbol{\Theta}(\boldsymbol{x},t)$ be a smooth coordinate transformation of $\delta\boldsymbol{x}$ to $\delta\boldsymbol{z}$, i.e., $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$. Then, a symmetric continuously differentiable matrix $\boldsymbol{M}(\boldsymbol{x},t) := \boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t)$ is said to be a *metric* of the system (1).

Definition 3.3. The covariant derivative of $\boldsymbol{f}(\boldsymbol{x},t)$ in $\delta\boldsymbol{x}$ coordinate is represented as

$$\boldsymbol{F} := \left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} + \boldsymbol{\Theta}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)\boldsymbol{\Theta}^{-1},$$

and is called the *generalized Jacobian*. This can be easily derived by differentiating $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$ with respect to t, leading to $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{z} = \boldsymbol{F}\boldsymbol{z}$.

3.1.2 Fundamental Theorem of Contraction Theory

The following theorem presents the fundamental theorem of contraction theory and corresponding necessary and sufficient condition for exponential convergence of the differential system (2).

Theorem 3.1 (see, T. 2.1 [3])**.** *If $\exists \boldsymbol{M}(\boldsymbol{x},t) = \boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t) > 0, \forall \boldsymbol{x}, t$ where $\boldsymbol{\Theta}(\boldsymbol{x},t)$ defines a smooth coordinate transformation of $\delta\boldsymbol{x}$ to $\delta\boldsymbol{z}$, i.e., $\delta\boldsymbol{z} = \Theta(\boldsymbol{x},t)\delta\boldsymbol{x}$, subject to the following equivalent conditions holds for $\exists \alpha \in \mathbb{R}_{>0},\ \forall \boldsymbol{x}, t$:*

$$\begin{aligned}\lambda_{\max}(\boldsymbol{F}(\boldsymbol{x},t)) = \lambda_{\max}\left(\left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} + \boldsymbol{\Theta}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)\boldsymbol{\Theta}^{-1}\right) &\leq -\alpha,\\ \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + \mathrm{sym}(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}) &\leq -2\alpha\boldsymbol{M},\end{aligned}$$

where the arguments of $\boldsymbol{M}(\boldsymbol{x},t)$ and $\boldsymbol{F}(\boldsymbol{x},t)$ are omitted for simplicity, then, the system (1) is said to be contracting with an exponential rate α, i.e., all trajectories of (1) converge to a single trajectory. The converse is also true.

Proof. Taking time derivative of a differential Lyapunov function of $\delta\boldsymbol{x}$, $V = \delta\boldsymbol{z}^\top\boldsymbol{z} = \delta\boldsymbol{x}^\top\boldsymbol{M}\delta\boldsymbol{x}$, using the differential dynamics (2), we have

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}V(\boldsymbol{x},\delta\boldsymbol{x},t) =& \delta\boldsymbol{x}^\top\left(\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + \mathrm{sym}(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}})\right)\delta\boldsymbol{x} = 2\delta\boldsymbol{z}^\top\boldsymbol{F}\delta\boldsymbol{z}\\ \leq& -2\alpha\delta\boldsymbol{x}^\top\boldsymbol{M}\delta\boldsymbol{x} = -2\alpha\delta\boldsymbol{z}^\top\delta\boldsymbol{z} = -2\alpha V.\end{aligned}$$

According to the comparison lemma (Lemma **??**), we have $V(t) \leq V(0)e^{-2\alpha t}$, which then yields $\|\delta\boldsymbol{z}(t)\|^2 \leq \|\delta\boldsymbol{z}(0)\|^2 e^{-2\alpha t}$ and $\|\delta\boldsymbol{z}(t)\| \leq \|\delta\boldsymbol{z}(0)\|e^{-\alpha t}$. This implies that any infinitesimal displacement $\delta\boldsymbol{z}$ and $\delta\boldsymbol{x}$ converges to zero exponentially with rate α. Note that the initial conditions are exponentially "forgotten" as time goes on. The proof of the converse can be found in [2, Sec. 3.5]. □

It is notable that the unboundedness of the metric $\boldsymbol{M}(\boldsymbol{x},t)$ does not create any problem in a technical sense. This is because, the dynamics of $\boldsymbol{M}(\boldsymbol{x},t)$ is linear with infinite escape time. Therefore, it can be handled by renormalizing the metric $\boldsymbol{M}(\boldsymbol{x},t)$.

Theorem 3.1 can also be proven by using the transformed squared length integrated over two arbitrary solutions of (1). The following theorem presents the alternative proof of Theorem 3.1.

Theorem 3.2 (see, T. 2.3 [3])**.** *Let $\boldsymbol{\xi}_1(t)$ and $\boldsymbol{\xi}_2(t)$ be two solutions of (1), and define the transformed squared length with $\boldsymbol{M}(\boldsymbol{x},t)$ of Theorem 3.1 as follows:*

$$V_{sl}(\boldsymbol{x},\delta\boldsymbol{x},t) = \int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \|\delta\boldsymbol{z}\|^2 = \int_0^1 \frac{\partial \boldsymbol{x}}{\partial \mu}^\top \boldsymbol{M}(\boldsymbol{x},t)\frac{\partial \boldsymbol{x}}{\partial \mu}\mathrm{d}\mu, \tag{4}$$

where $\boldsymbol{x}$ is a smooth path parameterized as $\boldsymbol{x}(\mu=0,t)=\boldsymbol{\xi}_1(t)$ and $\boldsymbol{x}(\mu=1,t)=\boldsymbol{\xi}_2(t)$ by $\mu\in\{0,1\}$. Also, define the path integral with the transformation $\boldsymbol{\Theta}(\boldsymbol{x},t)$ for $\boldsymbol{M}(\boldsymbol{x},t)=\boldsymbol{\Theta}(\boldsymbol{x},t)^\top\boldsymbol{\Theta}(\boldsymbol{x},t)$ as follows:

$$V_l(\boldsymbol{x},\delta\boldsymbol{x},t) = \int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \|\delta\boldsymbol{z}\| = \int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \|\boldsymbol{\Theta}(\boldsymbol{x},t)\delta\boldsymbol{x}\|. \tag{5}$$

Then, (4) and (5) satisfy the following inequality:

$$\|\boldsymbol{\xi}_1(t)-\boldsymbol{\xi}_2(t)\| = \|\int_{\boldsymbol{\xi}_1(t)}^{\boldsymbol{\xi}_2(t)} \delta\boldsymbol{x}\| \le \frac{V_l}{\sqrt{\underline{m}}} \le \sqrt{\frac{V_{sl}}{\underline{m}}},$$

where $\boldsymbol{M}(\boldsymbol{x},t)\ge \underline{m}\boldsymbol{I}_n$, $\forall \boldsymbol{x},t$ for $\exists \underline{m}\in\mathbb{R}_{>0}$ and Theorem 3.1 can also be proven by using (4) and (5) as a Lyapunov-like function, resulting in incremental exponential stability of the system (1) (see, Definition 3.1). Note that the shortest path integral V_l of (5) with a parameterized state $\boldsymbol{x}$ (i.e., $\inf V_l = \sqrt{\inf V_{sl}}$) defines the Riemannian distance and the path integral of a minimizing geodesic.

Proof. ... □

Example 3.1. Consider the following system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & x_1 \\ -x_1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

However, aforementioned theorems are limited to the convergence of the state trajectory to a single trajectory. In the next section, we introduce the partial contraction theory to investigate the convergence of the state trajectory to a desired trajectory.

3.1.3 Partial Contraction

[6,7]

3.2 Basic Results of Contraction Theory for Stochastic Systems

4 Conclusion

References

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